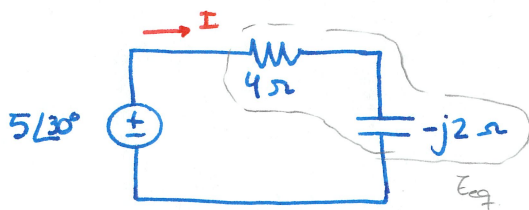


EXERCISE 2 (Average power)

Find average power supplied by source and absorbed by resistor



Average power is calculated as follows: $P = \frac{1}{2} \operatorname{Re}[VI^*] = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$

Also, for resistors: $P = \frac{1}{2} V_m I_m = \frac{1}{2} R |I|^2 = \frac{1}{2} \frac{|V|^2}{R}$

First, calculate current phasor I :

$$I = \frac{5\angle 30^\circ}{Z_{eq}} = \frac{5\angle 30^\circ}{4 - j2} = \frac{5\angle 30^\circ}{4.47\angle -27.57^\circ} = 1.118\angle 56.57^\circ \text{ A}$$

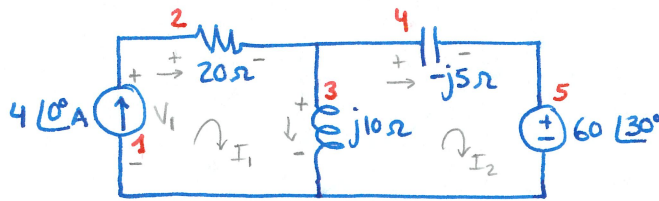
$$\underline{P_{\text{source}}} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} \times 5 \times 1.118 \times \cos(30 - 56.57^\circ) = \underline{2.5 \text{ W}} \text{ (supplied)}$$

$$\underline{P_{4\Omega}} = \frac{1}{2} R |I|^2 = \frac{1}{2} \times 4 \times 1.118^2 = \underline{2.5 \text{ W}} \text{ (absorbed)}$$

Conservation of energy
(since $P_{j2} = 0 \text{ W}$)

EXERCISE 4 (Average power)

Determine the average power for all elements



$$I_1 = 4\angle 0^\circ \text{ A}$$

$$\text{KVL in mesh 2: } -j10(I_1 - I_2) + (-j5)I_2 + 60\angle 30^\circ = 0$$

$$-j10I_1 + j10I_2 - j5I_2 = -60\angle 30^\circ$$

$$-j40 + j5I_2 = -60\angle 30^\circ \Rightarrow j5I_2 = 60\angle -150^\circ + j40$$

$$I_2 = \frac{60\angle -150^\circ + 40\angle -90^\circ}{5\angle 90^\circ} = \frac{-51.96 - 30j + j40}{5\angle 90^\circ} = \frac{-51.96 + j10}{5\angle 90^\circ} = \frac{52.91\angle 169.11^\circ}{5\angle 90^\circ}$$

$$I_2 = 10.58\angle 79.11^\circ = 2 + j10.39$$

* Power of current source

$$P_{I_g} = -\frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$

passive sign convention $\rightarrow$ KVL in mesh 1 to calculate V_1 :

$$-V_1 + 20I_1 + j10(I_1 - I_2) = 0$$

$$V_1 = 20I_1 + j10(I_1 - I_2); \quad V_1 = 80 + j10(4 - 2 - j10.39) = 80 + j10(2 - j10.39)$$

$$V_1 = 80 + j20 - j^2 103.9 = 183.9 + j20 = 184.98\angle 6.21^\circ$$

$$\underline{P_{I_g}} = -\frac{1}{2} \times 184.98 \times 4 \times \cos(6.21^\circ - 0^\circ) = \underline{\underline{-367.79 \text{ W}}}$$

* Power of voltage source

$$\underline{P_{V_g}} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} \times 60 \times 10.58 \times \cos(30^\circ - 79.11^\circ) = \underline{\underline{207.8 \text{ W}}}$$

* Power of circuit elements:

$\rightarrow$ The voltage source is dissipating power

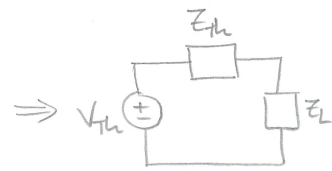
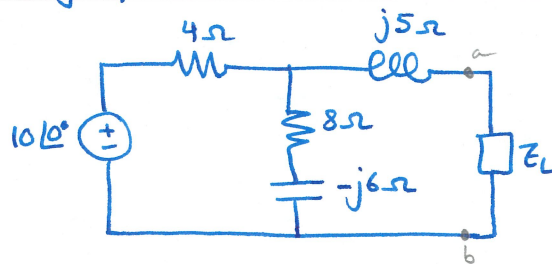
$$P_L = P_C = 0$$

$$\underline{P_R} = \frac{1}{2} R I_m^2 = \frac{1}{2} \times 20 \times 4^2 = \underline{\underline{160 \text{ W}}}$$

$$\Sigma P = 0 \quad (\text{conservation of energy})$$

EXERCISE 6 (Max. average power transfer)

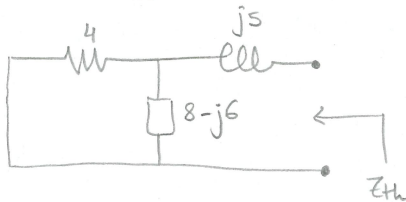
Find Z_L that maximizes the average power from the circuit below and find the maximum average power



MAPT $\Rightarrow Z_L = Z_{Th}^*$

$$P_{max} = \frac{|V_{Th}|^2}{8 R_{Th}}$$

* First, calculate Z_{Th} :



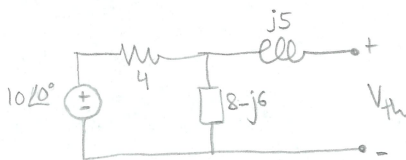
$$Z_{Th} = j5 + [4 \parallel (8 - j6)]$$

$$Z_{Th} = j5 + \frac{4 \times (8 - j6)}{4 + 8 - j6} = j5 + \frac{32 - j24}{12 - j6} = j5 + \frac{40 \angle -36.87^\circ}{13.42 \angle -26.57^\circ}$$

$$Z_{Th} = j5 + 2.98 \angle -10.3^\circ = j5 + 2.93 - j0.53$$

$$Z_{Th} = 2.93 + j4.47 \Rightarrow \underbrace{Z_L = 2.93 - j4.47}_{\substack{\uparrow \\ R_{Th}}}$$

* Then, calculate V_{Th} :



$$V_{Th} = V_{oc} = V_{8-j6} = 10 \frac{8 - j6}{4 + 8 - j6} = \frac{80 - j60}{12 - j6} = \frac{100 \angle -36.87^\circ}{13.42 \angle -26.57^\circ}$$

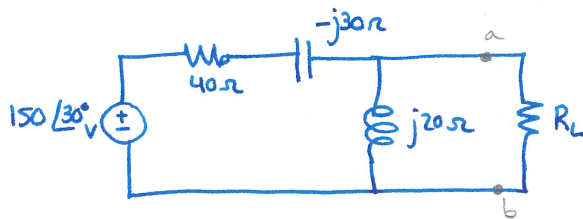
voltage divider

$$V_{Th} = 7.45 \angle -10.3^\circ$$

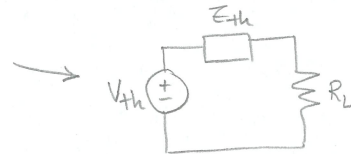
* Calculate P_{max} :

$$\underline{P_{max} = \frac{|V_{Th}|^2}{8 R_{Th}} = \frac{7.45^2}{8 \times 2.93} = 2.368 \text{ W}}$$

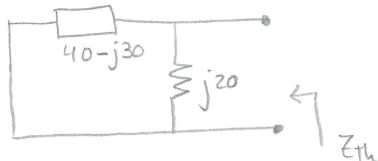
EXERCISE 8 (Max. average power transfer)



Find R_L for max. average power and then calculate that power



* Calculate Z_{th} :



MAPT $\Rightarrow R_L = |Z_{th}|$
for resistive load

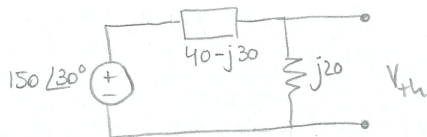
$$P_{max} = \frac{1}{2} R_L |I_L|^2 = \frac{1}{2} \frac{|V_L|^2}{R_L}$$

$$Z_{th} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{40 - j30 + j20} = \frac{j800 - j^2 600}{40 - j10}$$

$$Z_{th} = \frac{60 + j80}{4 - j} = \frac{100 \angle 53.13^\circ}{4.12 \angle -14.04^\circ} = 24.27 \angle 67.17^\circ$$

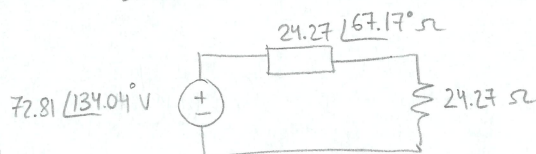
$$R_L = 24.27 \Omega$$

* Calculate V_{th} :



$$V_{th} = V_{oc} = V_{j20} = 150 \angle 30^\circ \times \frac{j20}{40 - j30 + j20} = \frac{150 \angle 30^\circ \times 20 \angle 90^\circ}{40 - j10} = \frac{300 \angle 120^\circ}{4.12 \angle -14.04^\circ} = 72.81 \angle 134.04^\circ$$

voltage divider



* Calculate P_{max} :

$$P_{max} = \frac{1}{2} R_L |I_L|^2 = \frac{1}{2} \frac{|V_L|^2}{R_L}$$

$$V_L = 72.81 \angle 134.04^\circ \times \frac{24.27}{24.27 + 24.27 \angle 67.17^\circ} = \frac{1767.1}{24.27 + (9.42 + j22.37)} = \frac{1767.1}{33.69 + j22.37} = \frac{1767.1}{40.44 \angle 33.58^\circ}$$

voltage divider

$$V_L = 43.7 \angle -33.58^\circ \text{ V}$$

$$P_{max} = \frac{1}{2} \times \frac{43.7^2}{24.27} = 39.34 \text{ W}$$